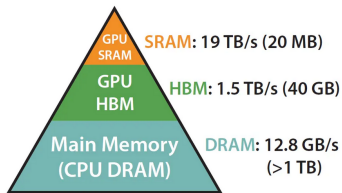


Deep Learning for NLP

Training LLMs

How to reduce compute and memory



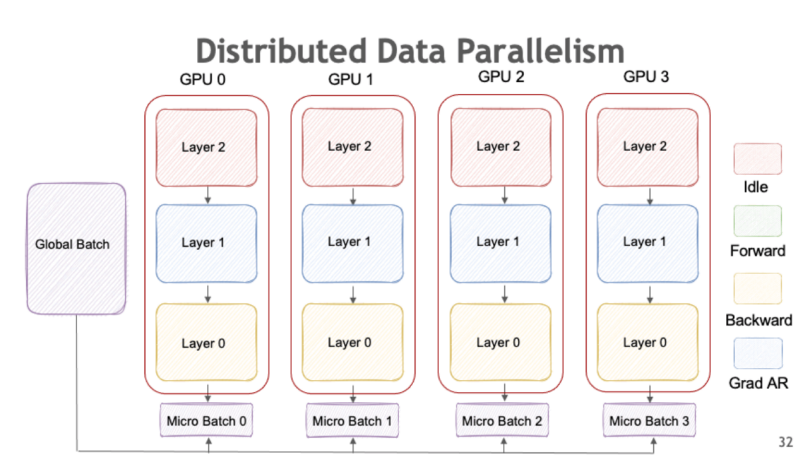
Learning goals

- Learn about different techniques to reduce compute and memory
- Learn about distributed training with data/tensor parallelism
- Learn about FlashAttention

DISTRIBUTED TRAINING

- Training LLMs faster on many GPUs
- Avoiding OOM (out of memory) issues
- **Data parallelism:** split the data on different model replicas
- **Tensor parallelism:** split model parameters accross GPUs

DATA PARALLELISM (1) (ANIMATED GIF!)



Source: Nvidia

DATA PARALLELISM (2)

- **Data Splitting:**

- The dataset is divided into smaller chunks, and each chunk is assigned to a different processing unit (e.g., GPU or CPU) on different nodes
- Each node processes a different subset of the data in parallel, reducing the overall training time

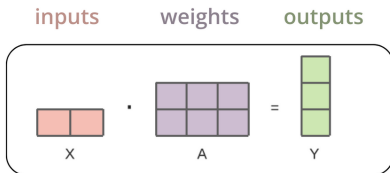
- **Model Replication:**

- Each processing unit has a replica of the neural network model
- These replicas are trained independently on their respective data subsets

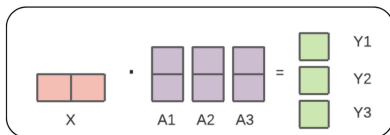
- **Gradient Aggregation (All Reduce):**

- After each forward and backward pass, gradients are computed locally on each node

TENSOR PARALLELISM (1)



is equivalent to



Source: Nvidia

TENSOR PARALLELISM (2)

- **Model Partitioning:**

- The model's layers or tensors are split across multiple devices
- Different parts of the model are assigned to different devices, enabling them to work on separate portions of the computations simultaneously

- **Forward and Backward Passes:**

- During the forward pass, each device processes its portion of the tensors with intermediate results passed between devices
- In the backward pass, gradients are computed in the reverse order, again with necessary data transfers between devices

- **Parameter Updates:**

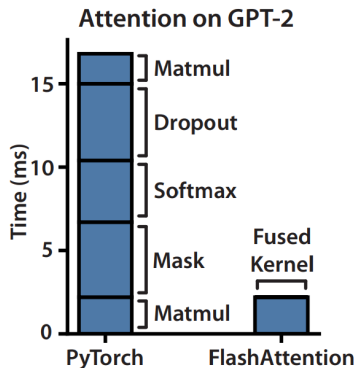
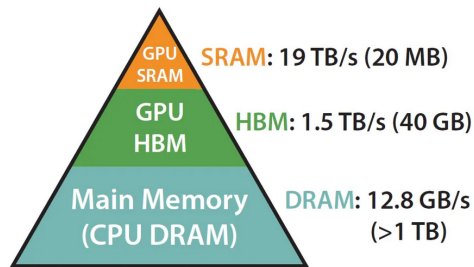
- Parameter updates can be performed independently on each device for the parameters they own
- After each update step, the devices synchronize to ensure consistency across the distributed model

FlashAttention

Fast and Memory-Efficient Exact Attention with IO-Awareness

- Fast
 - 15 % faster than BERT
 - 3x faster than GPT-2
 - 2.4x faster than Megatron-LM
- Memory-efficient
 - Reducing from $O(n^2)$ to $O(n)$
- Exact
 - Same as “vanilla attention”, not an approximation
- IO aware
 - Reducing memory load/store operations

GPU MEMORY HIERARCHY



► Source: Dao, 2022

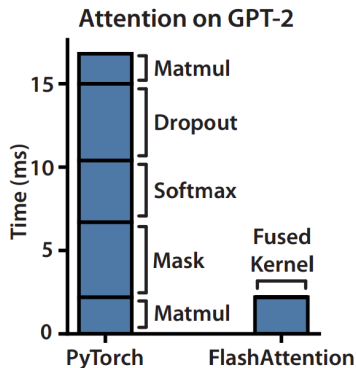
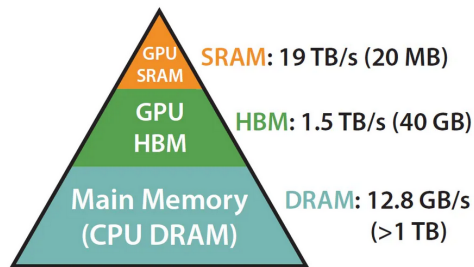
COMPUTING CONSIDERATIONS

- GPU compute has been growing faster than memory bandwidth
 - GPU has to wait for data
- Transformer operations are memory-bound
 - Elementwise operations with high memory access
- IO aware means reducing memory load/store operations

FLASHATTENTION

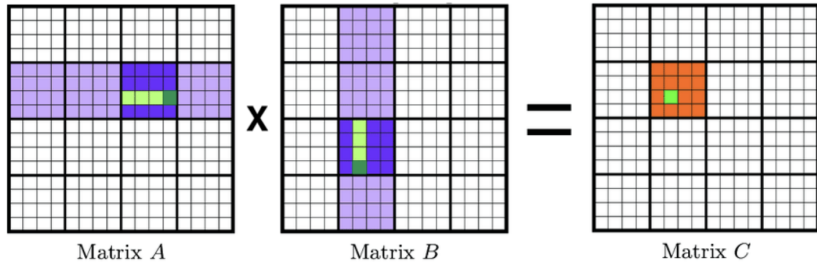
- Tiling or chunking of matrices into blocks
- Operation fusion
 - For example: first matrix multiplication (query-key), softmax, second matrix multiplication (weight-value)
- This minimizes redundant swapping in and out of data
- General idea: you avoid that the large intermediate tensors (e.g., pre-softmax) are written to HBM and then read back into (SRAM) memory

SRAM VS HBM / FUSED KERNEL

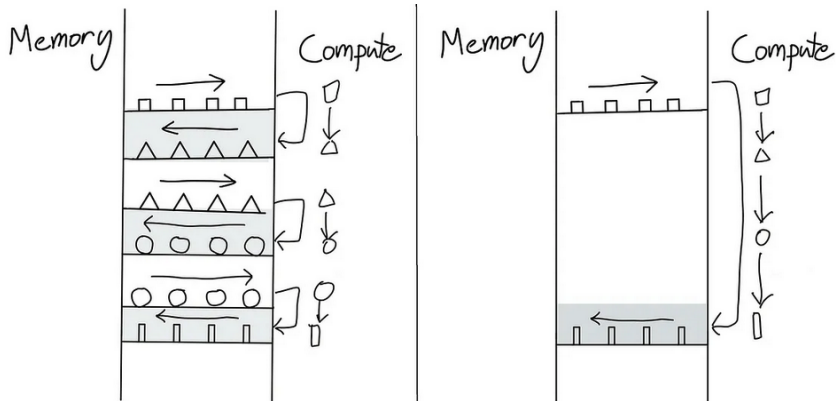


► Source: Dao, 2022

TILING



OPERATION FUSION



Source: https://horace.io/brrr_intro.html

LIMITATIONS AND PROSPECTS

- FlashAttention requires writing attention to CUDA language
 - A new CUDA kernel for each new attention implementation
 - CUDA is lower-level than PyTorch
 - Implementation may not be transferable accross GPUs
- Towards IO-Aware Deep Learning
 - Extending beyond attention
 - DeepSeek
- Multi-GPU IO-Aware Methods
 - FlashAttention computation may be parallelizable accross multiple GPUs